(Please write your Enrollment Number) Enrollment No. [redacted]

MID-TERM EXAMINATION
(Course Name: B.Tech.) (Semester: 1)
(October 2024) OFF LINE mode

Subject Code: BAI 104	Subject: Programming Fundamentals
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)
a)	Define what an algorithm is. Write a simple algorithm to calculate the sum of two numbers.	
b)	Create a simple Python class named *Car* with attributes *make*, *model*, and *year*. Create an object from this class and print the object's details.	
c)	Given the following list, analyze and explain the output of the following code: my_list = [2, 4, 6, 8, 10] print(my_list[1:4])	
d)	List at least three built-in methods for strings in Python. Provide a brief explanation of each method.	
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**

a)	Given the following class structure, evaluate how inheritance and method overriding are used. Write a short program to demonstrate this:
	class Vehicle:
	def start(self):
	return "Starting the vehicle"
	
	class Car(Vehicle):
	def start(self):
	return "Starting the car"	

b)	Write a Python program that uses an if-else structure and logical operators to determine if a given year is a leap year. Explain the conditions used.	
c)	Evaluate the pros and cons of using loops versus recursion for solving problems in Python. For which kind of problems would recursion be a better choice? Provide a Python example to support your evaluation.	
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
a)	Given a string, write a Python function to check if it is a palindrome. Explain how string indexing and slicing are used in the solution.	
b)	Explain the concept of string slicing and how it allows manipulation of substrings in Python. How does slicing work with negative indices? Provide examples to showcase the use of slicing for both positive and negative indices.	
c)	Write a Python function that takes a tuple of numbers and returns the sum of all elements in the tuple. How would the function differ if you were to pass a list instead of a tuple? Discuss the immutability of tuples in your explanation. def sum_of_tuple(t): return sum(t) numbers_tuple = (1, 2, 3, 4, 5) print(sum_of_tuple(numbers_tuple))	